

Method overview to be redrawn.
 Body-Anchored Metric Calibration $\rightarrow$ Dynamic Interaction Grounding

Figure 2: GroundedHuman reads body–scene relations for two complementary corrections. Metric queries calibrate scene depth and camera translation against human anchors. Given this reference, interaction attention selects evidence around the current body and converts it into a whole-body translation update. Recomputing the evidence makes grounding responsive to the changing placement.

3 METHOD

Given a monocular video $\mathcal{I} = \{\mathbf{I}_t\}_{t=1}^T$, GroundedHuman recovers scene geometry, camera motion, and human meshes in a shared metric world coordinate system. Our central idea is to use the human body as a *structured interface to the scene*: its metric extent provides a reference for scene calibration, while its spatial configuration determines which scene observations are relevant to human placement. We encode these body–scene relations through queries and attention, and read out the geometric corrections they support. Interaction is represented implicitly in these relations and their induced updates, without an intermediate contact-label prediction.

This principle gives rise to two modules (Fig. 2). **Body-Anchored Metric Calibration** uses *Metric Alignment Query* to relate human anchors to scene features and recover a shared metric reference. **Dynamic Interaction Grounding** uses *Placement-Conditioned Interaction Attention* to infer translation corrections from calibrated geometry around the body. Their ordering addresses an ambiguity: the same local depth discrepancy may reflect a scene-scale error or a misplaced human. We first calibrate the common reference, then interpret the remaining discrepancies as evidence for placement refinement. As the body moves, the attention context is rebuilt at its updated location.

Initial observations. Frozen VGGT- Ω provides relative Z-depth D_t^r , intrinsics $\mathbf{K}_t$, extrinsics $(\mathbf{R}_{cw,t}, \mathbf{t}_{cw,t}^r)$, and multi-layer scene features $\mathbf{F}_t$. Under the same intrinsics, frozen NLF provides metric SMPL observations: pose θ_{tq} , shape β_{tq} , root translation $\tau_{tq}^{(0)}$, and confidence c_{tq} for person q . Let $\bar{\mathbf{V}}_{tq}$ denote the posed vertices before root translation; the current camera-space body is $\mathbf{V}_{tq}^{c,(k)} = \bar{\mathbf{V}}_{tq} + \tau_{tq}^{(k)}$, with translation broadcast over vertices. These estimates supply the human metric prior; pose and shape are preserved throughout both modules.

3.1 BODY-ANCHORED METRIC CALIBRATION

Metric calibration asks how scene geometry should be expressed relative to the human metric reference. Surface correspondences initialize this relation; structured body queries then read scene features to estimate the remaining correction.

Human-surface initialization. Projecting an initial body vertex into D_t^r yields a pair of depths along the same camera ray. We pool valid depth ratios across people:

$$s_t^c = \text{median}_{(q,i) \in \mathcal{A}_t} \frac{[\mathbf{V}_{tqi}^{c,(0)}]_z}{D_t^r \left(\pi(\mathbf{K}_t, \mathbf{V}_{tqi}^{c,(0)}) \right)}, \quad D_t^{\text{coarse}} = s_t^c D_t^r. \quad (1)$$

Here, π is perspective projection and $\mathcal{A}_t$ retains valid correspondences after projection and depth filtering, keeping the nearest sampled human surface when projections coincide. The median reduces sensitivity to isolated errors. Sampling and missing-observation handling are detailed in Appendix C.

Metric Alignment Query. An anchor specifies both *where* to retrieve scene evidence and *how* to interpret it relative to the body. We construct 24 anchor tokens per person from the body state, anchor position and projection, and a nearby observation from D_t^{coarse} : its 3D position, normal, anchor-relative offset, distance, and depth residual. A 3×3 neighborhood of $\mathbf{F}_t$ supplies scene tokens $\mathbf{S}_{tqj}$

for anchor j . Self-attention first combines evidence across body anchors; each anchor then queries its scene neighborhood. Suppressing frame and person indices, the attention read is

$$\begin{aligned}\tilde{\mathbf{H}}^{(l)} &= \mathbf{H}^{(l)} + \text{SelfAttn}\left(\text{LN}(\mathbf{H}^{(l)})\right), \\ \mathbf{Q}_j &= \text{LN}(\tilde{\mathbf{h}}_j^{(l)})\mathbf{W}_Q, \quad \mathbf{K}_j = \mathbf{S}_j\mathbf{W}_K, \quad \mathbf{U}_j = \mathbf{S}_j\mathbf{W}_V, \\ \mathbf{o}_j &= \text{softmax}\left(\frac{\mathbf{Q}_j\mathbf{K}_j^T}{\sqrt{d}}\right)\mathbf{U}_j.\end{aligned}\tag{2}$$

The final line shows one attention head of dimension d ; $\mathbf{U}_j$ denotes scene values to distinguish them from body vertices. Multi-head outputs, residual connections, and a feed-forward block update the anchor tokens. Pooling their final responses predicts residual log-scale $\Delta\ell_{tq}$ and depth bias Δb_{tq} . Thus the query reads scene information conditioned on a body–scene discrepancy, with the analytic initialization accounting for the dominant scale mismatch.

Shared metric readout. Normalized NLF confidences combine person-level predictions into frame-level residual scale ρ_t and bias b_t (Appendix C). Assuming a common scene scale within the input clip, we aggregate

$$\begin{aligned}s^{\text{vid}} &= \exp\left[\text{median}_t \log(s_t^c \rho_t)\right], \quad b^{\text{vid}} = \text{median}_t b_t, \\ D_t^m &= s^{\text{vid}} D_t^r + b^{\text{vid}}, \quad \mathbf{t}_{cw,t}^m = s^{\text{vid}} \mathbf{t}_{cw,t}^r.\end{aligned}\tag{3}$$

Only depth receives the additive bias; camera rotation and intrinsics are unchanged. Back-projecting D_t^m with $\mathbf{K}_t$ gives camera-space metric points $\mathbf{P}_t^m$, the reference for subsequent placement refinement.

3.2 DYNAMIC INTERACTION GROUNDING

With the metric reference established, grounding asks how a body’s placement should change in response to its surroundings. Moving the body changes nearby observations, their geometric residuals, and their relevance. We therefore construct a placement-dependent interaction representation, read out a translation update, and recompute the representation at the revised location.

Placement-Conditioned Interaction Attention. We partition the SMPL surface into 96 corresponding regions, each represented by its centroid and eight sampled vertices. At iteration k , each region gathers scene points from four supports, $\mathcal{S} = \{\mathcal{W}_3, \mathcal{W}_7, \mathcal{W}_{\text{adp}}, \mathcal{W}_{\text{ann}}\}$: fixed 3×3 and 7×7 windows, an adaptive window following its projected extent, and an outer annulus. These supports expose fine alignment cues together with broader environmental structure.

Each sample is encoded relative to the current body region using 3D and image-plane offsets and depth residuals. Comparison with an instance-aware human depth buffer separates observations associated with the person’s own surface from environmental evidence. Other-human samples are excluded from both attention channels and retained only as interference statistics. Let $\mathbf{z}_{ri}^{(k)}$ encode sample i relative to region r , and $\mathcal{N}_{rsc}^{(k)}$ be the valid sample set for support $s \in \mathcal{S}$ and channel $c \in \{\text{self}, \text{env}\}$. Masked attention forms

$$\mathbf{u}_{rsc}^{(k)} = \sum_{i \in \mathcal{N}_{rsc}^{(k)}} \frac{\exp\left(a(\mathbf{z}_{ri}^{(k)})\right)}{\sum_{j \in \mathcal{N}_{rsc}^{(k)}} \exp\left(a(\mathbf{z}_{rj}^{(k)})\right)} \mathbf{z}_{ri}^{(k)},\tag{4}$$

where a is a learned scalar score; an empty set returns a zero token. Four supports and two channels yield eight context tokens per region, accompanied by valid-support ratios. Both the encoded relations and sampled sets depend on the current placement. Consequently, the same scene can yield different interaction evidence as the body moves.

From interaction evidence to placement. Context tokens are fused with region geometry, body-region embeddings, support ratios, and interference statistics to obtain $\mathbf{h}_r^{(k)}$. Each region predicts a

bounded translation proposal $\mathbf{v}_r^{(k)}$, gate $g_r^{(k)}$, and scalar log-variance $\ell_r^{(k)}$. Proposals are expressed along the region’s viewing direction and two tangent directions, then converted to camera coordinates. Each proposes a translation of the *whole body*. With $m_r^{(k)}$ indicating valid support, we aggregate

$$\alpha_r^{(k)} = m_r^{(k)} g_r^{(k)} \exp(-\ell_r^{(k)}), \quad \hat{\mathbf{v}}^{(k)} = \frac{\sum_r \alpha_r^{(k)} \mathbf{v}_r^{(k)}}{\max\left(\sum_r \alpha_r^{(k)}, \varepsilon\right)}. \quad (5)$$

The readout favors supported, reliable evidence. A person-level gate controls the bounded root update:

$$\begin{aligned} \Delta \tau_{tq}^{(k)} &= g_{tq}^{\text{person},(k)} \text{clip}_{[-d_{\max}, d_{\max}]^3}(\hat{\mathbf{v}}_{tq}^{(k)}), \\ \tau_{tq}^{(k+1)} &= \tau_{tq}^{(k)} + \Delta \tau_{tq}^{(k)}. \end{aligned} \quad (6)$$

Two feed-forward steps share parameters and rebuild region locations, sample ownership, and attention context after each update. Here, *dynamic* refers to this placement-dependent recomputation. Scene geometry, camera parameters, body pose, and shape remain fixed. Interaction is learned through the displacement supported by body–scene relations, without explicit contact classification or test-time optimization.

World-space reconstruction. Extrinsics follow $\mathbf{X}^c = \mathbf{R}_{cw,t} \mathbf{X}^w + \mathbf{t}_{cw,t}^m$. The final human meshes are

$$\mathbf{V}_{tq}^w = \mathbf{R}_{cw,t}^\top \left(\bar{\mathbf{V}}_{tq} + \tau_{tq}^{(2)} - \mathbf{t}_{cw,t}^m \right), \quad (7)$$

applied vertexwise. The same inverse transform maps $\mathbf{P}_t^m$ to world coordinates, placing humans and scene geometry in a common metric frame.

3.3 TRAINING OBJECTIVES

The two stages are trained separately, with VGGT- Ω and NLF frozen. Metric calibration uses metrically supervised sequences with synthetic scale perturbations. Initial scale training is followed by training on analytically initialized states, supervising the residual correction required from that state:

$$\mathcal{L}_{\text{calib}} = \lambda_s \mathcal{L}_{\text{scale}} + \lambda_b \mathcal{L}_{\text{bias}} + \lambda_d \mathcal{L}_{\text{depth}}. \quad (8)$$

The terms supervise residual log-scale, depth bias, and calibrated depth. In the residual-head training stage, the anchor interaction backbone is frozen and only the scale and bias heads are updated. The bias target is zero for this scale-perturbation path, constraining additive compensation for multiplicative errors.

With calibration fixed, Dynamic Interaction Grounding is trained on clean, scale-perturbed, translation-perturbed, and mixed states:

$$\mathcal{L}_{\text{ground}} = \lambda_v \mathcal{L}_{\text{vote}} + \lambda_t \mathcal{L}_{\text{trans}} + \lambda_g \mathcal{L}_{\text{gate}} + \lambda_r \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{state}}. \quad (9)$$

Regional votes target the remaining displacement divided by the number of remaining updates. Final-translation supervision trains the placement readout, gate supervision trains update validity, and regularization controls update magnitude. State-dependent terms encourage recovery from large errors, preservation of accurate inputs, and non-degrading corrections across iterations. Appendix D details the objectives and training configuration status.